Byte 2 Beat Hackathon

Cardiovascular Disease Prediction: Data Analysis and Model Development

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Introduction

Cardiovascular disease remains one of the leading causes of mortality worldwide, and early detection is critical yet unevenly distributed across populations. Machine learning offers a path toward scalable, data-driven screening tools that can support clinical decision-making without requiring specialist access. This project addresses two complementary clinical scenarios: population-level risk screening using basic health examination data, and diagnostic confirmation using advanced stress test and ECG measurements.

Datasets and Cleaning

The cardiac dataset contains demographic, physiological, and lifestyle features including age, blood pressure, cholesterol, glucose, and binary lifestyle flags. The heart dataset contains clinical measurements at rest and during exercise stress tests, with categorical variables pre-encoded as booleans. Both use a binary disease target.

Both datasets required removal of biologically implausible values. For the cardiac dataset, blood pressure, BMI, and age were filtered to clinically acceptable ranges, reducing the dataset from 70 000 to 68 642 records. Age was also converted from days to years, and gender encoding was confirmed via average height. The heart dataset was filtered on equivalent criteria, reducing it from 918 to 746 records. Both targets remained balanced after cleaning (about 50/50), making accuracy a reliable criterion.

Exploratory Data Analysis

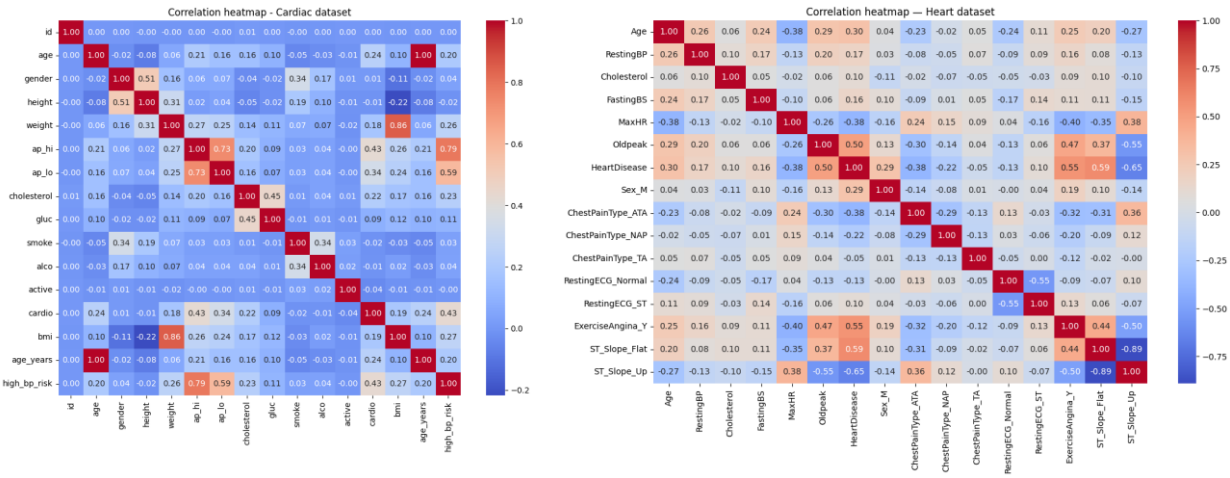

Figure 1. Correlation heatmaps for cardiac (left) and heart (right) datasets

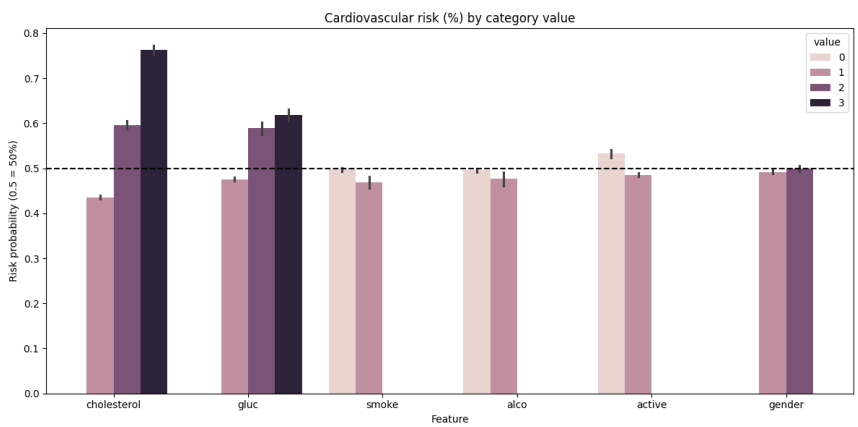

Figure 2. Cardiovascular disease risk rate by categorical feature value in the cardiac dataset

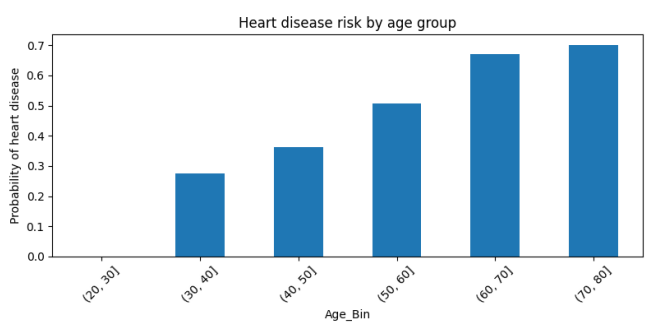

Figure 3. Heart disease risk probability by age group in the heart dataset

In the cardiac dataset, cholesterol was the strongest categorical predictor, with disease rates rising from 43.5% at level 1 to 76.3% at level 3 (Fig 2). Age and BMI showed clear risk thresholds: patients over 50 had a 54.6% disease rate versus 35.6% for those under 50, and obese patients (BMI higher than 30) showed 62.5% versus 44.9%. T-tests confirmed that all continuous features differed significantly between groups ($p < 0.0001$), with systolic BP showing the strongest linear correlation with the target ($r = 0.43$) (Fig 1).

Smoking and alcohol showed near-zero correlations and negligible group differences despite chi-square significance, suggesting a dataset composition effect rather than a genuine biological finding.

In the heart dataset, four features exceeded r = 0.5 with the target: ST_Slope_Up (-0.6), ST_Slope_Flat (0.59), ExerciseAngina_Y (0.55), and Oldpeak (0.5) (Fig 1), which are all exercise-derived features, confirming that direct measurements of cardiac function under stress are more discriminative than at rest or demographic features. Disease probability roughly doubled across age groups, from 35% for patients under 50 to 70% for those over 60 (Fig 3). Male patients showed a 34% higher disease rate than females, likely reflecting sampling characteristics of the clinical cohort rather than a pure biological difference.

Model Development

All models used an 80/20 train-test split with StandardScaler fitted on training data only. Hyperparameters were tuned via GridSearchCV (Random Forest) and RandomizedSearchCV (XGBoost). A feed-forward ANN (64-32-2, dropout 0.2, early stopping) was also built. The heart dataset was not subjected to ANN or RandomizedSearchCV due to its small size.

Model	Cardiac accuracy	Heart accuracy
Logistic regression (baseline)	72.4%	89.3%
Random forest	73.1%	90.7%
XGBoost	73.5%	88%
ANN	73.3%	N/A
Best CV score	73.5%	86.8%

Table 1. Model performance comparison across cardiac and heart datasets

The cardiac dataset had an accuracy plateau near 73% across all architectures, suggesting a performance ceiling imposed by feature quality rather than model complexity. The heart dataset consistently outperformed, as its features directly capture cardiac function under stress compared to demographic traits. The final selected models were a tuned XGBoost for cardiac and a Random forest for heart.

Evaluation

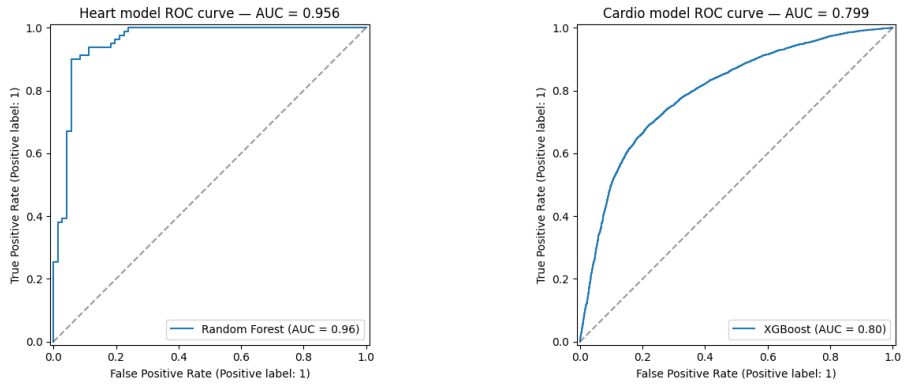

Figure 4. ROC curves for the heart model (left) and cardiac model (right)

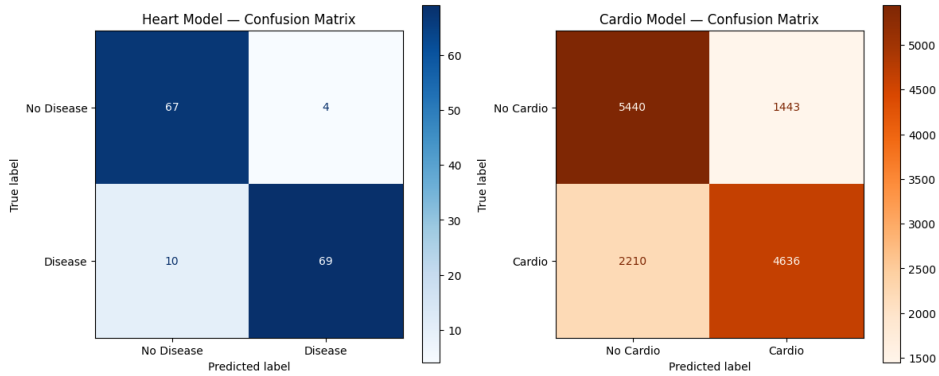

Figure 5. Confusion matrices for the heart model (left) and cardiac model (right)

The heart model achieved an AUC of 0.956, with its ROC curve reaching a 90% true positive rate at under 10% false positive rate already (Fig 4). The cardiac model achieved an AUC of 0.8 but struggled over 60% of healthy patients to catch 90% of sick ones (Fig 4), limiting its practical utility. Both models produced notable false negatives at the default threshold (Fig 5), but given the stong AUC scores, this probably reflects a calibration issue than a fundamental model failure.

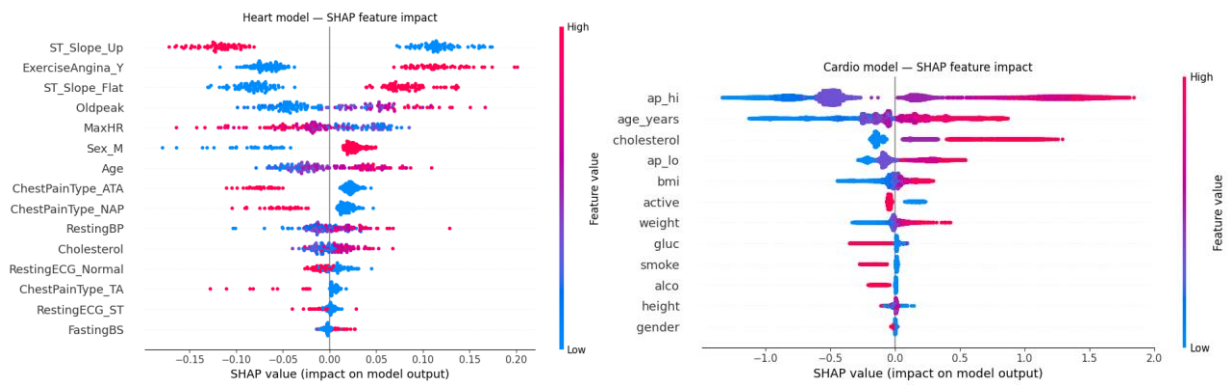

Figure 6. SHAP summary plots for the heart model (left) and cardiac model (right)

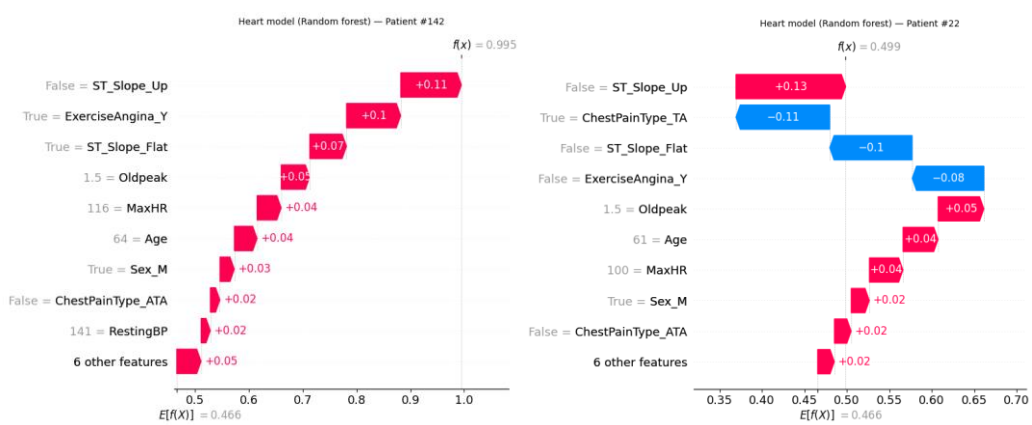

Figure 7. SHAP waterfall plots for confident (left) and uncertain (right) heart model predictions

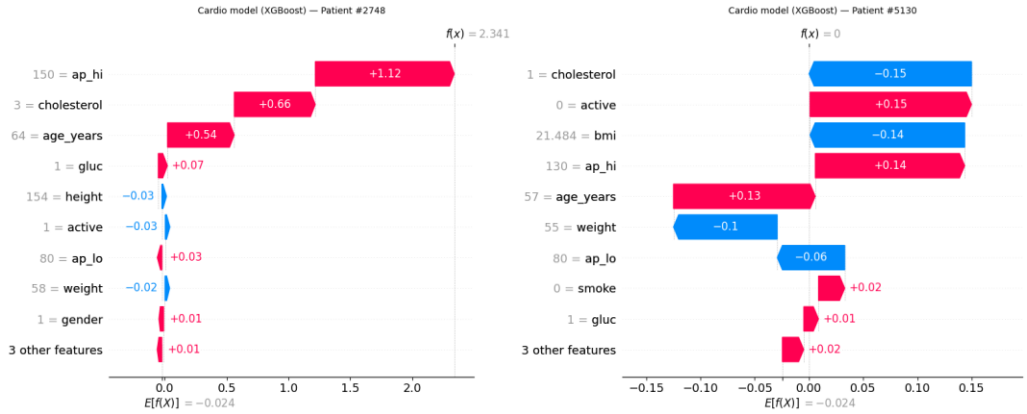

Figure 8. SHAP waterfall plots for confident (left) and uncertain (right) cardiac model predictions

SHAP analysis confirmed clinically coherent feature learning in both models. In the heart model, ST_Slope_Up consistently reduced predicted risk while ST_Slope_Flat, ExerciseAngina_Y, and high Oldpeak increased it (Fig 6), directly mirroring diagnostic criteria for heart disease. In the cardiac model, systolic BP, age, and cholesterol dominated as expected (Fig 6), though a notable bias emerged: absolute weight carried higher SHAP magnitude than BMI, suggesting the model did not fully account for body proportionality. Waterfall plots further showed that confident predictions were driven by features aligned in one direction, resulting in a large gap between base value and final output, while uncertain predictions reflected genuinely balanced opposing signals rather than model error (Fig 8).

Conclusion

The tuned XGBoost on the cardiac dataset (73.5% accuracy, AUC of 0.8) suits broad population screening, while the Random forest on the heart dataset (90.7% accuracy, AUC of 0.96) suits clinical confirmation in symptomatic patients. The performance gap reflects feature quality rather than dataset size: demographic information overlaps heavily between classes, whereas exercise-derived cardiac measurements are much more discriminative. SHAP analysis confirmed interpretable, clinically grounded reasoning in both models. The primary remaining limitations are the false negative rate at the default threshold, which should be recalibrated toward higher recall before any clinical use, and the heart model's small sample size, which requires validation on a larger independent cohort. Future work could enrich the cardiac dataset with stress test features, combine both models into a two-stage screening pipeline, or validate on geographically diverse populations.

AI disclosure: Claude and Gemini were used as coding assistants for debugging and algorithmic suggestions. All analysis and interpretations were written by the author.